

## Problem Set 11

### More Complex Substitutions

September 29, 2026

#### Problem 1

(c)

$$\begin{aligned}
 u &= \ln x \\
 du &= \frac{1}{x} dx \\
 \int \frac{\ln^2 x}{x} dx &= \int u^2 du = \frac{u^3}{3} + C = \boxed{\frac{\ln^3 x}{3} + C}
 \end{aligned}$$

(d) Both are correct.

#### Problem 2

(e)

$$\begin{aligned}
 u &= \frac{3x^2}{\sqrt{2}} \\
 du &= 3\sqrt{2}x dx \\
 \int \frac{x}{2+9x^4} dx &= \frac{1}{6\sqrt{2}} \int \frac{1}{1+u^2} du \\
 &= \frac{1}{6\sqrt{2}} \arctan u + C = \boxed{\frac{1}{6\sqrt{2}} \arctan \frac{3x^2}{\sqrt{2}} + C}
 \end{aligned}$$

(g)

$$\begin{aligned}
 u &= \frac{1}{3} \tan 4x \\
 du &= \frac{4}{3} \sec^2 4x dx \\
 \int \frac{\sec^2 4x}{9 + \tan^2 4x} dx &= \frac{1}{12} \int \frac{1}{1+u^2} du \\
 &= \frac{1}{12} \arctan u + C = \boxed{\frac{1}{12} \arctan \frac{\tan 4x}{3} + C}
 \end{aligned}$$

#### Problem 3

(b)

$$u = x + 1$$

$$\begin{aligned}
 du &= dx \\
 \int x \sqrt[3]{x+1} \, dx &= \int (u-1) \sqrt[3]{u} \, du = \int u^{\frac{4}{3}} - u^{\frac{1}{3}} \, du \\
 &= \boxed{\frac{3}{7}(x+1)^{\frac{7}{3}} + \frac{3}{4}(x+1)^{\frac{4}{3}} + C}
 \end{aligned}$$

**Problem 4**

(c)

$$\int_0^{\frac{1}{4}} \sin^2 \pi x \, dx = \int_0^{\frac{1}{4}} \frac{1 - \cos 2\pi x}{2} \, dx = \boxed{\frac{1}{8} - \frac{1}{4\pi}}$$

(d)

$$\begin{aligned}
 u &= \cos x \\
 du &= -\sin x \, dx \\
 \int \sin^3 x \, dx &= \int (1 - \cos^2 x) \sin x \, dx \\
 &= \int u^2 - 1 \, du = \frac{u^3}{3} - u + C = \boxed{\frac{\cos^3 x}{3} - \cos x + C}
 \end{aligned}$$

(f)

$$\begin{aligned}
 u &= \frac{\pi}{2} - x \\
 du &= -dx \\
 \int_0^{\frac{\pi}{2}} \sin^2 x \, dx &= \int_0^{\frac{\pi}{2}} \sin^2 \left( \frac{\pi}{2} - x \right) \, dx = \int_0^{\frac{\pi}{2}} \cos^2 x \, dx
 \end{aligned}$$

**Problem 6**

(a)

$$\begin{aligned}
 u &= 4 - x \\
 du &= -dx \\
 \int_0^3 \frac{1}{\sqrt{4-x}} \, dx &= \int_1^4 \frac{1}{\sqrt{u}} \, du = 2\sqrt{u} \Big|_1^4 = \boxed{2}
 \end{aligned}$$

(b)

$$\begin{aligned}
 u &= \frac{x}{2} \\
 du &= \frac{dx}{2} \\
 \int_0^1 \frac{1}{\sqrt{4-x^2}} \, dx &= \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-u^2}} \, du = \arcsin \frac{1}{2} = \boxed{\frac{\pi}{6}}
 \end{aligned}$$

(c)

$$\begin{aligned}
 u &= \frac{x^2}{3} \\
 du &= \frac{2}{3} dx \\
 \int_0^{\sqrt{3}} \frac{x}{9+x^4} dx &= \frac{1}{6} \int_0^1 \frac{1}{1+u^2} du = \frac{1}{6} \arctan 1 = \boxed{\frac{\pi}{24}}
 \end{aligned}$$

(d)

$$\begin{aligned}
 u &= 9+x^2 \\
 du &= 2x dx \\
 \int_0^2 \frac{2x}{9+x^2} dx &= \int_9^{13} \frac{1}{u} du = \boxed{\ln \frac{13}{9}}
 \end{aligned}$$

(e)

$$\begin{aligned}
 u &= \arctan 2x \\
 du &= \frac{2}{1+4x^2} dx \\
 \int_0^{\frac{1}{2}} \frac{\arctan 2x}{1+4x^2} dx &= \int_0^{\frac{\pi}{4}} \frac{u}{2} du = \boxed{\frac{\pi^2}{64}}
 \end{aligned}$$

(f)

$$\begin{aligned}
 u &= \sqrt{x} \\
 du &= \frac{1}{2\sqrt{x}} dx \\
 \int_0^{\frac{\pi^2}{16}} \frac{\tan \sqrt{x}}{\sqrt{x}} dx &= \int_0^{\frac{\pi}{4}} 2 \tan u du = \int_0^{\frac{\pi}{4}} \frac{2 \sin u}{\cos u} du \\
 t &= \cos u \\
 dt &= -\sin u du \\
 \int_0^{\frac{\pi}{4}} \frac{2 \sin u}{\cos u} du &= \int_{\frac{\sqrt{2}}{2}}^1 \frac{2}{t} dt = 2(-\ln \frac{\sqrt{2}}{2}) = \boxed{\ln 2}
 \end{aligned}$$

**Problem 7**

(a)

$$x = \frac{1}{t}$$

$$dx = -\frac{1}{t^2} dt$$

$$\int_{\frac{1}{a}}^a f(x) \frac{dx}{x} = -\int_a^{\frac{1}{a}} f\left(\frac{1}{t}\right) \frac{dt}{t} = \int_{\frac{1}{a}}^a f\left(\frac{1}{x}\right) \frac{dx}{x}$$

(b)

$$2 \int_{\frac{1}{2}}^2 \frac{\ln x}{\frac{1}{x} + x} \frac{dx}{x} = \int_{\frac{1}{2}}^2 \left( \frac{\ln x}{\frac{1}{x} + x} + \frac{\ln \frac{1}{x}}{1 + \frac{1}{x^2}} \right) \frac{dx}{x} = \int_{\frac{1}{2}}^2 \left( \frac{\ln x - \ln x}{\frac{1}{x} + x} \right) \frac{dx}{x} = 0$$

$$\boxed{\int_{\frac{1}{2}}^2 \frac{\ln x}{1 + x^2} dx = 0}$$

(c)

$$2 \int_{\frac{1}{2}}^2 \arctan x \frac{dx}{x} = \int_{\frac{1}{2}}^2 \left( \arctan x + \arctan \frac{1}{x} \right) \frac{dx}{x}$$

$$= \int_{\frac{1}{2}}^2 \left( \arctan x + \frac{\pi}{2} - \arctan x \right) \frac{dx}{x} = \frac{\pi}{2} \int_{\frac{1}{2}}^2 \frac{dx}{x} = \frac{\pi}{2} (\ln 2 - \ln \frac{1}{2}) = \pi \ln 2$$

$$\int_{\frac{1}{2}}^2 \frac{\arctan x}{x} dx = \boxed{\frac{\pi}{2} \ln 2}$$

(d)

$$\int \frac{x^2 + 1}{x^4 + 1} dx = \int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$u = x - \frac{1}{x}$$

$$u^2 = x^2 + \frac{1}{x^2} - 2$$

$$du = 1 + \frac{1}{x^2} dx$$

$$\int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx = \int \frac{1}{2 + u^2} du$$

$$t = \frac{u}{\sqrt{2}}$$

$$dt = \frac{du}{\sqrt{2}}$$

$$\int \frac{1}{2 + u^2} du = \frac{1}{\sqrt{2}} \int \frac{1}{1 + t^2} dt = \frac{1}{\sqrt{2}} \arctan \frac{u}{\sqrt{2}} + C = \boxed{\frac{1}{\sqrt{2}} \arctan \left( \frac{x^2 - 1}{\sqrt{2}x} \right) + C}$$

(d) The value 0 is impossible, as the integrand is strictly greater than one on this interval. The error occurs as there is a jump discontinuity in the function's antiderivative.

$$\int_{-1}^1 \frac{x^2 + 1}{x^4 + 1} dx = 2 \int_0^1 \frac{x^2 + 1}{x^4 + 1} dx = 2(0 - \frac{\pi}{2\sqrt{2}}) = \boxed{\frac{\pi}{2}}$$